# AP Precalculus: 2026–2027

**Unit 4: Functions Involving Parameters, Vectors, and Matrices  Lesson 4.3: Parametric Functions and Rates of Change  (2 instructional periods, 55 minutes each)**

**Primary Objective:** Students will analyze the direction of planar motion from whether $x(t)$ and $y(t)$ are increasing or decreasing, compute average rates of change of each component over a parameter interval, and use their ratio to find the slope of the curve between two points. *(Big Idea: Functions)*

## Priority Ideas & Skills

| AP Skills | Key Understandings (EKs) |
|---|---|
| **3.B** *Apply numerical results* — Compute the average rate of change (ARC) of $x(t)$ and $y(t)$ over $[t_1, t_2]$ and apply the ratio $\Delta y/\Delta x$ to determine the slope of the parametric curve between two points. | **EK 4.3.A.1**: As the parameter increases, direction of planar motion can be analyzed in terms of $x$ and $y$ independently. If $x(t)$ is increasing, the particle moves right; if decreasing, left. If $y(t)$ is increasing, the particle moves up; if decreasing, down. |
| **3.C** *Support conclusions with rationale/data* — Justify claims about direction of motion and verify that two different parametrizations of the same path yield the same secant slope. | **EK 4.3.A.2**: At any given point in the plane, the direction of planar motion may be different for different values of $t$. |
| | **EK 4.3.A.3**: The same curve can be parametrized in different ways and traversed in different directions with different parametric functions. |
| | **EK 4.3.A.4**: Over $[t_1, t_2]$, the ratio of the ARC of $y(t)$ to the ARC of $x(t)$ gives the slope of the curve between the two corresponding points, provided ARC of $x(t) \neq 0$. |

## Vocabulary, Concepts, & Theorems

| Term | Definition / Note |
|---|---|
| direction of motion | Determined by whether each component is increasing or decreasing: $x(t)$ increasing $\Rightarrow$ right; $y(t)$ increasing $\Rightarrow$ up. |
| ARC of $x(t)$ | $\dfrac{x(t_2) - x(t_1)}{t_2 - t_1}$; measures how fast $x$ changes per unit $t$. |
| ARC of $y(t)$ | $\dfrac{y(t_2) - y(t_1)}{t_2 - t_1}$; measures how fast $y$ changes per unit $t$. |
| slope of parametric curve | $\dfrac{\Delta y}{\Delta x} = \dfrac{\text{ARC of } y(t)}{\text{ARC of } x(t)}$ when ARC of $x(t) \neq 0$; equals the slope of the secant line connecting the two points on the curve. |
| reparametrization | A different pair of component functions that traces the same set of points in the plane, possibly in a different direction. |

## Activate Prior Knowledge & Spiral Review

**Spiral topics activated during Warm-Up (first 8 minutes):**

- **Increasing/decreasing functions**: Determine whether a function is increasing or decreasing on an interval from its formula or a table — directly used to identify direction of motion in each component.
- **Average rate of change**: Compute $\dfrac{f(b)-f(a)}{b-a}$ over an interval — the core calculation for both ARC of $x(t)$ and ARC of $y(t)$.
- **Parametric functions as two components**: The idea that a parametric function $f(t) = (x(t), y(t))$ outputs ordered pairs — prerequisite for analyzing each component independently.

## Hook — Entry Event

**Scenario:** Two drones follow the same figure-eight path over a field — Drone A clockwise, Drone B counter-clockwise. From below, both trace identical curves in the sky, yet at any given moment they head in opposite directions.

**Discussion prompts:**

1. *"Can the same path in the plane be described by two completely different parametric functions? What would have to be different about them?"*
2. *"What determines which way a drone is heading at any moment? Can you tell just from its position?"*
3. *"If you knew the drone's $x$-position was increasing, which horizontal direction is it moving?"*

**Key tensions to surface:** (1) Position alone does not determine direction — you need to know whether $x(t)$ and $y(t)$ are increasing or decreasing. (2) The same geometric curve can be traversed in opposite directions by different parametrizations.

**Transition:** "Today we develop two tools: reading direction of motion from the components, and computing the slope of the curve between two points using average rates of change."

## Lesson — Instruction Sequence

**Step 1 — Direction of Motion** (10 min)
Write $f(t) = (x(t), y(t))$. Emphasize: analyze $x$ and $y$ *independently*. If $x(t)$ is increasing over an interval, the particle moves **right**; if decreasing, **left**. If $y(t)$ is increasing, the particle moves **up**; if decreasing, **down**. Cover EK 4.3.A.1. Example: $f(t) = (t^2-2t, 3t-t^2)$. At $t = 0.5$: compare $x(0)$ and $x(1)$ — $x(0) = 0$, $x(1) = -1$, so $x$ decreased ⇒ moving **left**. Compare $y(0) = 0$ and $y(1) = 2$ — $y$ increased ⇒ moving **up**. Ask: "Does knowing the drone's $(x, y)$ position tell you which way it is heading?" (No — EK 4.3.A.2.)

**Step 2 — Same Curve, Different Directions** (8 min)
Cover EK 4.3.A.3. Example: compare $f(t) = (t, t^2)$ for $0 \le t \le 3$ and $g(t) = (3-t, (3-t)^2)$ for $0 \le t \le 3$. Build a small table for both: the ordered pairs are the same set of points, but traversed in opposite directions. Key question: "How would you verify that two parametric functions trace the same path?" (Show both produce the same $(x, y)$ values.) Stress: the curve is a geometric object; the parametrization adds direction and timing.

**Step 3 — Average Rates of Change** (10 min)
Define ARC of $x(t)$ over $[t_1, t_2]$: $\dfrac{x(t_2)-x(t_1)}{t_2-t_1}$, and similarly for $y(t)$. Then: slope of curve between the two points equals $\dfrac{\Delta y}{\Delta x}$, which can be computed as $\dfrac{\text{ARC of } y(t)}{\text{ARC of } x(t)}$. Cover EK 4.3.A.4. Caution: if ARC of $x = 0$, slope is undefined (vertical secant). Cold-call: "Does the ratio depend on $\Delta t$? Does it simplify to just $\Delta y/\Delta x$?"

**Step 4 — Full Example:** $f(t) = (t^2 - 2t,\ 3t - t^2)$ (12 min)
**Direction at** $t = 0.5$: $x(0) = 0$, $x(1) = -1$ (decreased) ⇒ moving left. $y(0) = 0$, $y(1) = 2$ (increased) ⇒ moving up.
**Direction at** $t = 2.5$: $x(2) = 0$, $x(3) = 3$ (increased) ⇒ moving right. $y(2) = 2$, $y(3) = 0$ (decreased) ⇒ moving down.
**Slope of secant from** $t = 1$ **to** $t = 3$: $x(1) = -1$, $x(3) = 3$, $\Delta x = 4$; $y(1) = 2$, $y(3) = 0$, $\Delta y = -2$; slope $= -2/4 = -1/2$.

**Pacing note:** Steps 1–2 in Period 1; Steps 3–4 + activity in Period 2.

## Active Monitoring

- **Confusing direction with sign of the component**: Students may say "$x(t) < 0$ so the particle is moving left." Prompt: *"Is $x(t)$ getting larger or smaller as $t$ increases? That tells you direction, not the sign of $x$ itself."*
- **Using $\Delta y/\Delta t$ as slope**: Remind students the slope of the *curve* uses $\Delta y/\Delta x$, not $\Delta y/\Delta t$. Cold-call: "Show me which two changes go in the numerator and denominator for the slope of the curve."
- **Assuming direction is constant on an interval**: Students may check only endpoints. Prompt: "Can the direction change inside the interval? How would you check at an interior value of $t$?"
- **Reparametrization confusion**: Students may think reversing the path gives a different curve. Ask: *"Do both parametrizations produce the same set of $(x, y)$ ordered pairs? Does the* slope *of the secant change?"*

## Group Work & Differentiation

Students work in groups of 3–4 on the tiered activity (assign by tier or allow self-selection with guidance). The activity uses two robots tracing the same parabolic path in opposite directions.

**Tier R — Remediate**
Given $f(t) = (3t, -2t + 6)$ on $0 \le t \le 4$:

- Determine direction (right/left, up/down) at $t = 1$.
- Compute ARC of $x(t)$ and ARC of $y(t)$ over $[0, 2]$.
- Find slope of the curve between $t = 0$ and $t = 2$.

**Tier A — Approaching**
Given $f(t) = (t^2 - 1, \sin(\pi t))$ on $0 \le t \le 2$:

- Identify intervals where particle moves right/left, up/down.
- Find slope of secant from $t = 0$ to $t = 2$.

**Tier E — Extend**
Write a second parametrization of the straight-line path traced by $h(t) = (3t, -2t + 6)$ on $0 \le t \le 4$ but traversed in the *opposite* direction. Verify the slope of the secant between the same two endpoints has the same value.

## Individual Work & Assessment

Exit ticket administered independently (final 5 minutes of Period 2).

1. A particle moves with $f(t) = (2t - t^2,\ t^2 - 4t + 3)$ for $0 \le t \le 4$. At $t = 0.5$, is the particle moving right or left? Up or down? Show your reasoning.
2. Compute the ARC of $x(t)$ on $[0, 3]$ and the ARC of $y(t)$ on $[0, 3]$. What is the slope of the curve between these two points?

**Data use:** Students who cannot determine direction from increasing/decreasing need to revisit Step 1. Students who report $\Delta y/\Delta t$ as the slope (instead of $\Delta y/\Delta x$) need the ratio clarified before Lesson 4.4.

## Reinforcement & Extension

**Homework (Lesson 4.3):** 5–6 problems covering direction of motion analysis, computing slope via the ARC ratio, a problem with two different parametrizations of the same straight line, and 1 AP-style multiple-choice item.
**Preview of Lesson 4.4:** Ask students: *"Today you found the slope of a secant between two points on a parametric curve using average rates of change. What happens to that slope as the two parameter values get closer and closer together? Could you find the slope of the* tangent *line?"* Lesson 4.4 connects average rates of change to instantaneous rates of change in the parametric context.